Project Report
Machine Learning & Language Models
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Abstract

Building on a coursework project comparing BERT representations with fMRI data, we investigate brain-activity predictivity across the Language-LH, Semantic-AAL, and Left Fusiform Face Area networks. We compare layer-wise neural predictivity with predictivity of sentence length and topic. Topic-prediction accuracy shows strong positive correlations with neural predictivity across all three regions, whereas sentence-length predictivity shows no significant correlations. These findings suggest an association between topic-related predictivity and brain-activity predictivity across BERT layers, although the correspondence between linguistic features and specific brain regions can not be confirmed.

1 Introduction

This report describes the work undertaken during the final project of the course Machine Learning & Language Models. The lecture gives an overview of essential Machine Learning algorithms and their relation to applications in Language Modelling, and the student's performance is assessed in the context of this project which aims at incorporating methods learnt during the course.

The dataset from Pereira et al., 2018 is the central component of the project. It contains fMRI scan data collected over three experimental settings laid out in the paper. For the purposes of the project, only experiment 2 is used, which has a total of 8 participants. All participants were asked to read 96 text passages, each containing four sentences, which span 24 topics. The fMRI scan data was collected in the form of voxel-level blood oxygenation changes, to measure the brain response to each of the 384 stimulus sentences.

The main focus of the project is comparing the data in the Pereira dataset to LLM representations, in this case of the model BERT. Bidirectional Encoder Representations from Transformers is an encoder-only bidirectional language model first presented in Devlin et al., 2019. It uses a Masked Language Modelling algorithm to build vector representations of token meaning, called embeddings, from its pre-training data. These embeddings may later be used for NLP tasks. In our case, they are compared to the fMRI and stimulus data from the Pereira dataset.

The first part of the project consists of a predetermined Python notebook section where the dataset is explored and initial comparisons made. The second part of the project is an extension on the notebook, in which we explore a novel research question, drawing on the methods and data established in the previous notebook. Each part of the project is described in its respective section.

2 Notebook

The first part of the notebook is concerned with importing data of one participant and creating a class that makes the data more manageable. After that, the 384 phrases are tokenised and given to the pretrained BERT model as an input, which allows us to access the embeddings of the phrases at each layer. These embeddings are pooled, in our case using the CLS token, so that every sentence is represented by only one token. In this way we can fit regression models or classifiers on the tokens of phrases at each layer, to predict brain activity in the left hemisphere language network or linguistics properties of the sentences such as sentence length and the topic of a sentence.

R^2 and accuracy results show that, while the sentence length feature is better captured in early layers of BERT, topic prediction, a high level linguistic feature, is better captured in later layers. As opposed to that, the Language-LH brain activity is better represented in middle layers, more precisely in the 5th one in the case of participant M02.

3 Extension

3.1 Research Topic

In our own extension we are interested in comparing different networks of the brain, and in particular, exploring wether the activity in networks with more abstract tasks such as the semantic Automated Anatomical Labelling network (S-AAL), are better predicted by layers of BERT that also capture more complex features of the stimuli, such as the topic of the sentence. Conversely, we expect low-level features such as sentence length to be encoded in layers of BERT that best capture brain activity of areas such as the Visual Word Form Area, responsible for recognising written letters. Since this area was not functionally localised in the case of the participants of the experiment, we use the Left Fusiform Face Area (L-FFA) to approximate the activity of the Visual Word Form Area, on the basis that both areas are part of the Fusiform Gyrus and overlap to an extent. An overview of the ROI's is provided in Table 1, which includes size as maximum voxel count as well as the level of linguistic processing at which they contribute.

	L-FFA	**L LH**	**S AAL**
n voxels	≤ 639	≤ 5906	≤ 11480
Level	Low	General	High

Table 1: Overview of ROI's

3.2 Methods

For our analyses we use BERT layer embeddings from each of the Transformer architecture's 12 hidden layers, obtained in the notebook using the CLS pooling strategy. In contrast to the notebook, the extension makes use of data from all participants of experiment 2, which requires a slight modification of the data processing methods and the class provided to us in the notebook.

We measure brain predictivity from the embeddings in three Regions of Interest, namely the Left Fusiform Face Area, the Language Left Hemisphere network, and the Semantic-AAL network. All three regions are established separately as voxel subsets of the participants' brains in the Pereira data set and brain activity is predicted using a Ridge Regression model. The model is fitted layer-wise to the embeddings and sentence-wise fMRI data, making use of 5-fold cross-validation. Predictivity performance per participant and layer is assessed using the same metric as in the notebook, namely R^2, and layer-wise results obtained calculating the mean R^2 over all participants. This decision enables us to conduct layer-wise predictibility analysis across all ROI's. We carry out a permutation test with 1000 permutations to establish statistical significance of brain predictivity for each voxel, discarding insignificant voxels when computing the mean and plotting results.

In addition to our analysis of the Pereira dataset's results, we measure predictibility of sentence length and topic from the embeddings, similarly to the notebook. A Ridge Regression model is used for sentence length prediction and mean R^2 metrics used to assess performance. The topic feature is predicted with the help of a Logistic Regression model instead, to accommodate for the classification task, and the performance assessed with accuracy rather than R^2. In both cases, the mean over the 384 stimuli sentences is taken for each layer.

To compare predictivity of linguistic features and of brain activity in all three regions, we instantiate a Spearman test, taking into account all voxels, as opposed to only the significant ones, to avoid potential bias. We choose this particular statistical test in consultation with our teaching assistant, given that our aim is to assess whether there is a monotonic relationship of the predictivities across layers. Since we are comparing different kinds of metrics, this excludes a numerical comparison. We apply the Benjamini–Hochberg correction across the six correlation tests to control for false discovery rate. We are aware that the results should only be taken as indications of correlation, since the Spearman test assumes independence of the layers, which need not be the case for the BERT model.

3.3 Results

Our three ROI's show distinct mean R^2 scores. The L-FFA region displayes a cross-layer mean of 0.043 and a highest R^2 of 0.045 at layer 8. The Language-LH network shows a cross-layer mean R^2 of 0.047 and a peak mean of 0.053 at layer 5, while the Semantic-AAL network has a cross-layer mean R^2 of 0.047 and a highest R^2 of 0.054 at layer 5. Results for all ROI's are visualised in Figure 1.

Embedding predictibility of sentence length shows a mean R^2 of 0.550, with layer 3 being the most predictive of the feature with an R^2 of 0.900 and dropping scores toward later layers. Meanwhile, predictibility of topics averages an accuracy

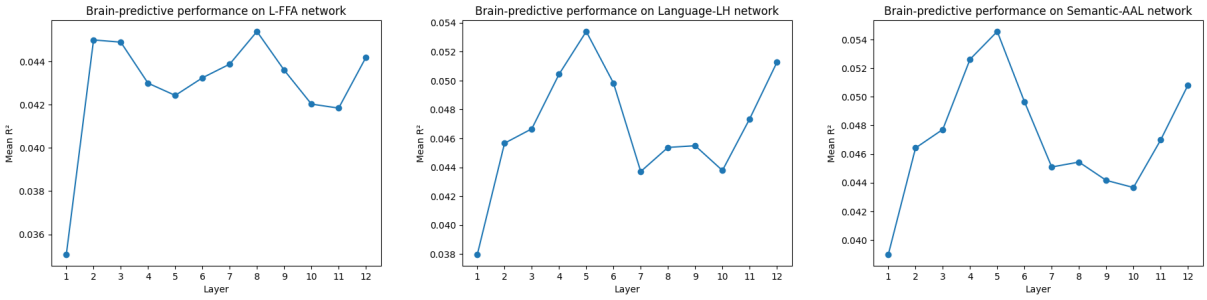


Figure 1: Predictibility from embeddings of betas across ROI's

of 0.704, peaking at 0.846 at layer 12. Contrary to sentence length, topics are clearly better predicted at later layers. Linguistic feature predictibilities are plotted in Figure 2.

Spearman's rank correlation analysis reveals strong positive correlations between sentence-topic predictivity and brain-activity predictivity across transformer layers in all three brain regions ($\rho_{\text{semantic}} = 0.818$ with $p = 0.0023$, $\rho_{\text{language}} = 0.881$ with $p = 0.0005$ and $\rho_{\text{visual}} = 0.937$ with $p = 4.196 \times 10^{-5}$; Benjamini-Hochberg correction is applied to all p-values). In contrast, no statistically significant correlations are found between sentence-length predictivity and brain-activity predictivity in any of the three regions after Benjamini–Hochberg correction for multiple comparisons.

4 Discussion

It is hard to say whether the results may partially confirm our hypothesis.

Immediately, the most striking observation is the shape of the Semantic-AAL curve compared to that of topic prediction accuracy. This is in line with our expectations. Both curves show a peak around layer 5, and increasing metrics towards later layers. In fact, the Spearman correlation is quite high between these two predictions and the corresponding p-value is significant. However, both curves also undeniably resemble that of the Language-LH network, and Spearman correlation of this network's predictivity with topic predictivity is very similar to that of Semantic-AAL. Most importantly, the brain-predictive performance on the Language-LH network and on the Semantic-AAL network show huge similarities in their graphs. This raises the question whether the distinction between general and high-level linguistic cognition is at all present in the brain predictivity of BERT embeddings, or wether topic prediction of sentences is as linked to the Semantic-AAL network as we hypothesised. Alternatively, brain response to such simple phrases might not meaningfully differentiate the Semantic-AAL network and the Language-LH network, making it hard to identify which one is most representative of topic predictivity from BERT embeddings.

As for the brain predictivity of the Left Fusiform Area, it seems comparatively uniform across layers, differently than sentence-length predictivity, which speaks against our initial assumptions. It is worth

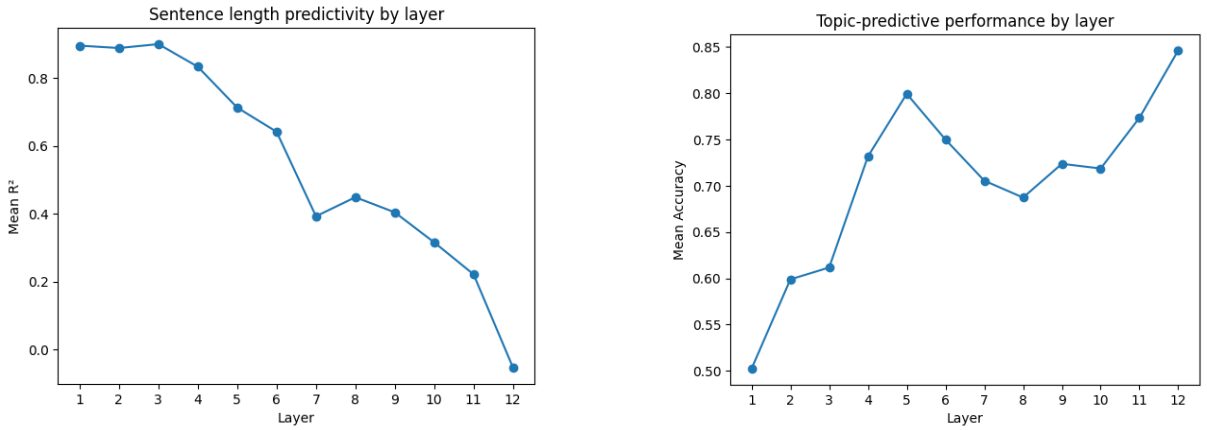


Figure 2: Predictibility of Linguistic features from embeddings

noticing that brain predictivity of this area largely differs from the Language-LH and the Semantic-AAL network. Still, this difference does not have a bearing on the contrastive comparison to linguistic features. There do not seem to be recognisable similarities between layer-wise sentence-length predictivity and Left Fusiform Area predictivity. Moreover, Spearman correlation for the L-FFA network is not significant in the case of this linguistic feature, while it is quite strong in the case of topic predictivity, like for the other brain regions.

In the aggregate, our results suggest that there may be some links between linguistic tasks like topic prediction and semantic brain networks as seen through the lens of BERT embeddings. Notably, this is clearly not a novel neurological discovery in itself, but rather a reflection of the potential architectural similarities and systematic localisation between BERT and human brains. It must be said, however, that this link seems to be present in all brain regions we explored, whereas no such connection can be found in the case of lower-level linguistic processing such as sentence-length. We must therefore admit that this is far from a definitive answer.

5 Limitations

The inconclusive outcome of our investigation may have several causes.

Firstly, our understanding of which network is best compared to which linguistic feature may not be sufficiently founded in neurological reality. The assumption that the length of a sentence might be best captured by the Fusiform Gyrus might be wrong, or the Left Fusiform Area might not be a good enough proxy for the Visual Word Form Area. Furthermore, even if the selected brain region can a priori be representative, the variation in the length of a sentence displayed to a participant need not be a strong enough signal to properly be displayed in said ROI.

Alternative stimuli, such as pictures, may be better suited for evaluating visual-linguistic brain regions. This may be done using VisualBERT or similar models to process the picture stimuli from the Pereira dataset's experiment 1, and later assessing brain predictivity on the picture embeddings of this model. Note that this would presuppose a similar architecture of BERT and the model used for picture processing, as otherwise no comparison between the brain predictivity of the two models across their layers can be made.

Secondly, a reason for the unclear results might be insufficient data. Experiment 2 only contained 8 participants, potentially obfuscating important patterns. Moreover, note that when inspecting the L-FFA network, we are only considering a single ROI, with at most 639 voxels, while the S-AAL network is composed of up to 11 thousand voxels. The size disparity surely has an effect on the quality and symmetry of the comparison, and makes for less representative results in the case of the L-FFA network.

Lastly, we again note that the Spearman correlation calculated can only be viewed as an approximate measure of correlation between the performances of predictions on the different layers, due to the potential dependence between BERTs layers.

6 Conclusions

In summary, our investigation of brain predictivity localisation and linguistic features within BERT's Transformer layer's saw mixed results. By comparing their positioning, we hoped to potentially contribute to a better understanding of BERT's brain-relational architecture. The results suggest that our hypothesis of brain-task correspondence in BERT's layers may potentially have some partial creedence. However, it was far from a doubtless connection, as the low-level features show essentially no correspondence, and a general vs. high-level task-localisation distinction is not detectable. Future research on the topic may include a more refined neurological basis and exploration of alternative modalities, such as vision, as our question remains open.

References

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